Hyperspectral Image Classification Based on Pyramid Coordinate Attention and Weighted Self-Distillation

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***Abstract*— Attention mechanism-based hyperspectral image (HSI) classification algorithms typically extract spectral and spatial features by spectral attention and spatial attention network, respectively. However, these algorithms lack joint attention and ignore imbalanced samples, leading to insufficient information extraction. To address this problem, this article proposes a novel HSI classification algorithm based on the pyramidal coordinate attention and weighted self-distillation (PCA-WSD). To perform the joint attention of spectral and spatial features, the proposed PCA mechanism uses spectral attention to cope with the diverse spatial features. The PCA mechanism consists of two components. First, the spatial pyramid coordinate squeeze (SPCS) is designed to aggregate spatial features with local and global information. Then, the tailored spatial pyramid coordinate excitation (SPCE) adaptively enhances their informative spectral features for the obtained spatial features, realizing the joint attention to spectral–spatial features. Furthermore, considering the imbalance of samples, WSD is proposed. Specifically, weighted cross-entropy is integrated into WSD. Extensive experiments are evaluated on the four HSI benchmark datasets: Indian Pine (IP), Pavia University (UP), Kennedy Space Center (KSC), and Pavia Center (PC). Compared with the seven advanced algorithms, the experimental results of the proposed algorithm reveal superior classification performance, especially for the imbalanced samples. Our code is available at https://github.com/githubltqc/PCAWSD.**

***Index Terms*— Hyperspectral image (HSI) image classification, spatial pyramid coordinate excitation (SPCE), spatial pyramid coordinate squeeze (SPCS), weighted self-distillation (WSD).**

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I. Introduction

HYPERSPECTRAL image (HSI) classification aims to assign each pixel a semantic-level label [1], that is, predict the land cover label that each pixel belongs to [2]. Different from other remote sensing images [3], HSI contains hundreds or even thousands of narrow and contiguous spectral bands [4]. It greatly enriches the spectral–spatial features of geological objects, which is more conducive to accurate identification. Therefore, it has been widely used in precision agriculture [5], geological survey [6], military surveillance [7], and so on. However, HSI classification remains challenging due to its inherent spectral high-dimensionality, limited labeled samples, and category imbalance. Therefore, how to extract more discriminative features from small and imbalanced data still has significant research value.

Researchers have made great efforts to extract more discriminative features and achieved remarkable success in recent years. Traditional machine learning (ML)-based algorithms improve feature discrimination by manually designing feature extractors [8], [9]. Initially, researchers only took the HSI pixel as input and proposed the pixel-wise algorithm [10]. However, these algorithms often result in lots of salt-and-pepper noise, which significantly hinders the classification. Researchers introduce spatial context information and propose a patch-wise algorithm considering the spatial dependence between pixels. Precisely, the patches are cropped from the original HSI. These patches follow the First Law of Geography [11], assuming that the adjacent pixels share the same label with the central pixel. The classical patch-wise classification algorithms include the Gabor transform [12], wavelet transform [13], and morphological analysis [14], which consider both the spectral information and spatial context information of pixels. However, these ML-based algorithms still rely heavily on expert knowledge when selecting features and hyperparameters, and their feature representation capabilities are relatively limited.

Unlike ML-based algorithms, deep learning (DL)-based algorithms can automatically extract more discriminative features [15], [16]. Among them, the convolutional neural network (CNN) has become the most popular network by local connection and weight-sharing [17], [18]. In 2014, Chen et al. [19] were the first to introduce the CNN into HSI classification. Specifically, principal component analysis (PCA) [20] was used to reduce the dimension of the input data, and then

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the CNN was used to extract spectral–spatial features automatically. Zhong et al. [21] constructed a spectral–spatial residual network (SSRN) using two consecutive spectral and spatial residual blocks to extract rich spectral and spatial features. Xu et al. [22] designed a dual-branch network with residual blocks called the dual-channel residual network (DCRN). By extracting more discriminative features from the spectral and spatial channels, the robustness of the model is improved. Paoletti et al. [23] proposed a lightweight CNN model that can be deployed on systems with limited computing resources. This model combined the Ghost module [24] with a CNN-based HSI classification algorithm to reduce the computational complexity and achieve efficient classification. Gao et al. [7] introduced mixed deep convolution and proposed a multiscale residual network (MSRN), which explored features at different scales while reducing the number of network parameters. However, all the above methods treat different bands and adjacent pixels equally, ignoring their uneven contribution to the classification, dramatically hindering improving classification performance.

In recent years, inspired by the attention mechanism of the human visual system, numerous attention mechanism-based methods have emerged [25], [26]. These methods pay more attention to helpful information and suppress useless information, thus effectively alleviating the problem of uneven contributions of different HSI bands and adjacent pixels. Sun et al. [27] proposed a spectral–spatial attention network (SSAN) to extract discriminative spectral–spatial features in the HSI attention region. Specifically, the spectral–spatial network (SSN) constructed the spectral and spatial modules, respectively. And then, the self-attention module was embedded into the SSN. Zhu et al. [2] explored the residual spectral–spatial attention network (RSSAN) to adaptively enhance the spectral and spatial features with significant contributions. In detail, the spectral attention module (SeAM) and spatial attention module (SaAM) were first designed. And then, they were embedded into the SSRN by sequential stacking. Li et al. [28] proposed joint spatial–spectral attention network (JSSAN), in which spatial–spectral attention (S^2A) block used a spatial attention branch, spectral attention branch, and attention distribution branch to capture long-range interdependency of the spatial–spectral features. Feng et al. [29] proposed a symmetric convolution generative adversarial network based on collaborative learning and attention mechanism. By embedding the spectral–spatial hard attention model into the generator, the distribution of generated samples is closer to the real HSI. Zhang et al. [30] constructed spectral partitioning residual network (SPRN), in which a new SaAM, called a homogeneous pixel detection module (HPDM), was proposed to capture homogeneous pixels in the patch. In HDPM, cosine similarity was adopted as a criterion to measure the similarity between the center pixel and other pixels in the patch. Zhang et al. [31] presented the spectral–spatial semantic network (S^3Net), which used multiple attention mechanism branches to extract informative features from three perspectives: spectral, spatial, and semantic, respectively.

The methods mentioned above are patch-wise algorithms, which generally have the following problems. First, it is challenging to model long-term context for a patch with a fixed size. Second, there is considerable overlap between adjacent patches, which inevitably increases the computational complexity. Third, the classification performance is sensitive to the patch size, which is also laborious. More and more whole image-wise studies were proposed to overcome these drawbacks, which tend to feed the whole image directly into the network [32], [33]. Xu et al. [34] first treated the HSI classification task as the natural image segmentation task and developed a spectral–spatial fully convolutional network (SSFCN) with a pixel-to-pixel structure. Zheng et al. [35] proposed fast patch-free global learning (FPGA) based on the encoder–decoder framework. It adopted a global stochastic stratified strategy to transform training samples into stratified samples and then embedded spectral attention into the encoder to maximize the use of global spatial information. Zhu et al. [4] proposed spectral–spatial-dependent global learning (SSDGL) to capture more discriminative feature representation further. It integrated global convolutional long short-term memory (GCL) and global joint attention mechanism (GJAM). Shen et al. [36] proposed a full convolution network (FCN) with an efficient nonlocal module named ENL-FCN. In this framework, criss-cross attention (CCA) [37] was employed to aggregate the relationship of pixels in the criss-cross path.

All the above attention methods have exhibited excellent performance, whether patch-wise or whole image-wise. They can be roughly summarized into three ways to extract spectral–spatial attention features. The first way is to extract spectral and spatial features using spectral and spatial models, respectively, and then input the spectral–spatial features into the attention model. The second way is to use the spectral and spatial attention branches to extract the spectral and spatial attention features, respectively, and then fuse them. The third way is that the spectral and spatial attention models are connected sequentially. All of them lack joint consideration of spectral–spatial attention.

By designing an appropriate spectral–spatial feature extraction mechanism, more discriminative features can be extracted from limited samples, which can alleviate the problem of small samples. The knowledge distillation [38], [39] proposed in recent years transfers the knowledge from the complex teacher model to the simple student model, improving the model's performance and alleviating the problem of small samples. However, designing an appropriate teacher network is time-consuming and laborious. So, the researchers proposed the self-distillation [40] that uses a single network as both a teacher and student model. It makes a single network model improve performance through knowledge distillation in self-learning. Yue [41] proposed a self-supervised learning method with adaptive distillation, which generated the adaptive soft label of each unlabeled sample. By calculating the similarity between labeled and unlabeled samples, the probability that it belongs to each category can be measured. However, this method does not consider the sample imbalance problem in HSI.

To address the aforementioned challenge of insufficient feature extraction caused by lacking of joint attention to spectral–spatial features and ignoring imbalanced samples, this

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article proposes pyramid coordinate attention and weighted self-distillation (PCA-WSD). First, spatial pyramid coordinate squeeze (SPCS) aims to aggregate diverse spatial information, which combines the advantages of coordinate attention and spatial pyramid. Subsequently, spatial pyramid coordinate excitation (SPCE) is specifically designed to emphasize informative spectral features for the aggregated spatial features. Combining SPCS and SPCE, a PCA module (PCAM) is constructed, which embeds diverse spatial attention features into spectral attention and realizes the joint attention of spectral–spatial features. Next, by stacking PCAM in a concatenate manner, the PCA network (PCAN) is built to enhance the reuse of features. Finally, weighted self-distillation (WSD) is used to further exploit the latent information and alleviate the problem of sample imbalance, in which both the teacher and the student networks are PCAN. By training these two networks, the logits can generate. The soft label is the probability distribution of logits after softmax processing. By reducing the Kullback-Leibler (KL) divergence of the teacher and the student soft label, the student network can continuously fit the teacher network, thereby exploiting the latent information. Meanwhile, to alleviate the sample imbalance, a weighted softmax is introduced to calculate the loss of ground truth and student soft label so that the model pays more attention to the difficult samples to classify.

The main contribution of PCA-WSD can be summarized as follows.

1) An attention squeeze method is designed to aggregate diverse spatial information. It comprehensively utilizes the advantages of spatial pyramid pooling and coordinate attention, which can effectively represent local features with multiscale information and global features with positional information.
2) An attention excitation method is specifically designed for the spatial features aggregated by SPCS. Then, SPCS and SPCE are combined to construct PCAM, which can adaptively emphasize the informative spectral bands under diverse spatial information and further realize the joint attention of spectral–spatial features.
3) WSD is first employed for HSI classification. Both soft labels generated by the teacher network and ground-truth labels are used to guide the student network training. This method not only captures the latent knowledge of the image, but also alleviates the sample imbalance.

The rest of the article is organized as follows. Section II describes the structure of the PCA-WSD algorithm. Section III introduces the comparison and ablation experiments. Section IV concludes this article.

II. Proposed Method

Attention mechanism-based approaches usually design spectral and spatial attention networks, respectively. And then connect them sequentially or with multiple parallels. These approaches fail to consider the joint attention and sample imbalance, which makes the extracted features insufficient. To deal with these problems, this article introduces a novel HSI classification algorithm called PCA-WSD. The framework of PCA-WSD is illustrated in Fig. 1.

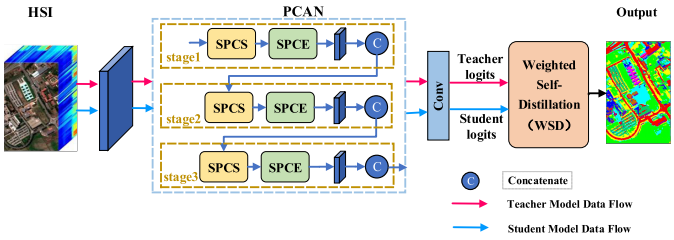


Fig. 1. Framework of PCA-WSD.

As shown in Fig. 1, the pink and blue lines represent the teacher and student data flow, respectively. The blue blocks represent convolution blocks, including batch normalization, 1×1 convolution, sigmoid, and 5×5 convolution. C indicates the cascade operation, which is cascaded with the PCAN input.

The PCA-WSD architecture consists of two branches, teacher and student, with pink and blue lines representing the data flow, respectively. Each branch uses PCAN and convolution blocks. Among them, PCAN contains stage 1, stage 2, and stage 3, which extract low, medium, and high-level features, respectively. Then, the multilayer features are fused by the residual connection between stages to achieve feature reuse. At each stage, SPCS and SPCE emphasize the spectral bands with a large amount of information under diverse spatial information to realize the joint attention of spectral and spatial features. Next, convolution blocks are used to enhance the nonlinearity of the network. Finally, the WSD strategy is adopted to guide students to explore potential information further and alleviate the problem of sample imbalance through teacher logit. Sections II-A–II-C describe the components of the PCA-WSD framework, respectively. Section II-D analyzes the computational complexity.

A. Spatial Pyramid Coordinate Squeeze

The existing algorithms introduce spectral and spatial attention to solve the problem of the uneven contribution of spectral bands and spatial pixels. However, they usually extract spectral and spatial attention features separately. The SeAM rarely considers spatial information, and the SaAM also ignores spectral information, which makes the extracted features inadequate.

In terms of spectral attention, most existing spectral attention first uses global pooling to aggregate spatial information and then uses spectral excitation to emphasize the informative band. Although this approach is simple, the informative band with a high local target response may have the same global mean pool (GAP) as the useless band in the noise background, which may cause ambiguity [42]. The spatial pyramid pool (SPP) can capture multiscale spatial information, but also use GAP to aggregate global features. Coordinate attention can extract global information with directionality to make up for the GAP. Therefore, this article fully uses the advantages of SPP and coordinated attention and proposes SPCS. The structure of the SPCS is shown in Fig. 2.

As shown in Fig. 2, the structure of SPCS mainly consists of pooling, flattening and reshaping, and concatenating. First,

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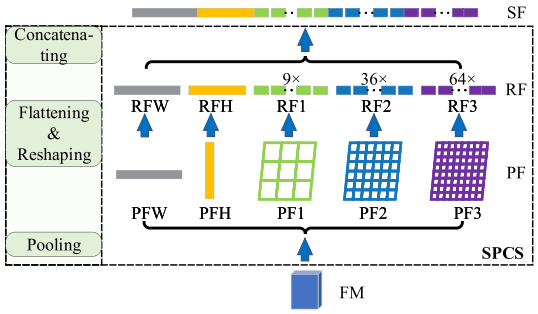


Fig. 2. Structure of SPCS.

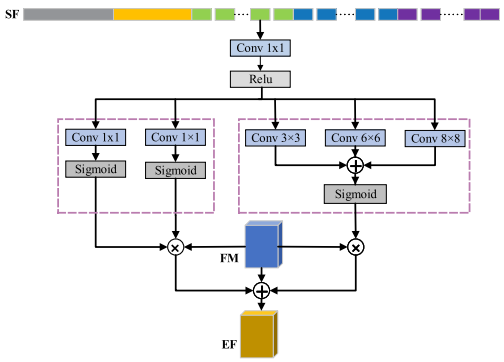


Fig. 3. Overall structure of SPCE.

the feature map FM is pooled to generate the PF. Specifically, the 1-D global pooling is performed on FM along the vertical and horizontal directions. After that, the FM is aggregated into two directional pooled feature vectors, PFW and PFH, which can capture long-range interactions. At the same time, the 2-D adaptive mean pooling operation is performed on FM. FM is aggregated into three feature maps, namely PF1, PF2, and PF3, with the sizes of 3×3, 6×6, and 8×8, which can extract local region information from different scales, greatly enriching the features' spatial information. Second, since the excitation in spectral attention only uses the channel vector feature as the input, the pooled features PF cannot be directly adopted. The PF is flattened and reshaped in this article to generate feature RF. Specifically, after this operation, RFW, RFH, RF1, RF2, and RF3 can be expressed as $f_r^w \in \mathbf{R}^{C \times H \times 1}$, $f_r^h \in \mathbf{R}^{C \times W \times 1}$, $f_r^1 \in \mathbf{R}^{C \times 9 \times 1}$, $f_r^2 \in \mathbf{R}^{C \times 36 \times 1}$, and $f_r^3 \in \mathbf{R}^{C \times 64 \times 1}$. The 9×, 36×, and 64× in Fig. 2 representing the number of image blocks is 9, 36, and 64, respectively. Finally, the feature RF is concatenated in the spatial dimension to generate the feature SF, denoted as $f_s \in \mathbf{R}^{C \times (H+W+S) \times 1}$, where $S = 9 + 16 + 64 = 109$. After the above operations, the spatial features with local and global information are aggregated, laying the foundation for the input of SPCE. Specifically, for the feature of the cth channel, the pooling operation is described as follows:

$$\begin{cases} f_{pc}^h(h) &= \dfrac{1}{W}\displaystyle\sum_{w=1}^{W} f_c(h,w) \\ f_{pc}^w(w) &= \dfrac{1}{H}\displaystyle\sum_{h=1}^{H} f_c(h,w) \\ f_{pc}^1 &= \text{AdaptiveAvgPool(3)} \\ f_{pc}^2 &= \text{AdaptiveAvgPool(6)} \\ f_{pc}^3 &= \text{AdaptiveAvgPool(8)} \end{cases} \tag{1}$$

where H and W are the height and width of the feature FM, respectively. $f_{pc}^h(h)$, $f_{pc}^w(w)$, f_{pc}^1, f_{pc}^2, and f_{pc}^3 denote the features of the cth channel of PFW, PFH, PF1, PF2, and PF3, which are generated by pooling FM. *A*daptiveAvgPool(3), *A*daptiveAvgPool(6), and *A*daptiveAvgPool(8) represent adaptive mean pooling with output sizes of 3×3, 6×6, and 8×8, respectively.

The cth channel of the feature RF is concatenated in the spatial dimension, as shown in (2)

$$f_{sc} = \left[f_{rc}^h, f_{rc}^w, f_{rc}^1, f_{rc}^2, f_{rc}^3\right] \tag{2}$$

where f_{sc} denotes the output feature of the cth channel after the SPCS, and [ ] denotes the concatenate operation in the spatial dimension. f_{rc}^h, f_{rc}^w, f_{rc}^1, f_{rc}^2, and f_{rc}^3 represent the features reshaped RFW, RFH, RF1, RF2, and RF3 in the cth channel, respectively.

B. Spatial Pyramid Coordinate Excitation

With SPCS, diverse spatial features with multiscale and positional information are aggregated. In this article, a tailored spectral excitation module, SPCE, is designed for these features to realize the joint attention of spectral–spatial features. Specifically, the features embedded with precise positional information are encoded as two attention maps. Each captures long-range dependencies along a specific direction, thereby preserving positional information in the generated spectral attention maps. In addition, features embedded with multiscale information are encoded as three attention maps of different scales, thereby preserving local features of different scales in the generated spectral attention maps. The overall structure of SPCE is shown in Fig. 3.

As seen in Fig. 3, the feature SF aggregated by SPCS is fed into SPCE to obtain the spectral–spatial attention feature. SPCE is mainly decomposed into two steps. First, $C \times 1 \times 1$ convolution is used to reduce SF's channel to $1/r$ of itself. Then, it is split into two sets from the spatial dimension, and the feature channel is expanded to C. By performing the above nonlinear operations on the channels, the weights of the channels with different spatial characteristics can be obtained. In turn, spectral attention features that fuse different spatial information can be captured. Specifically, the aggregated feature SF is first processed through 1×1 convolution and batch normalization. The calculation process is shown in (3)

$$f_m = \delta(F_1(f_s)) \tag{3}$$

where δ is the Relu activation function. F_1 represents the convolution operation and the kernel size is $C/r \times C \times 1 \times 1$, where C/r and C represent the number of output channels and input channels, respectively, and r represents the reduction ratio. f_s is the feature SF. After processing, the size of the MF becomes $C/r \times S \times 1$.

Then, a special channel expansion operation is designed. Specifically, features are split and reshaped along the spatial

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dimension and then divided into two groups. One group is the row-element MFH and column-element MFW, which contain positional information. The other features are MF1, MF2, and MF3, with sizes of 3×3, 6×6, and 8×8, containing multiscale information.

To capture the spectral attention features with positional information, for the row features MFH and column features MFW of size $(C/r, H, 1)$ and $(C/r, W, 1)$, two convolutions are used to automatically learn the importance of each channel in the positional features to obtain g^h and g^w. The calculation process is shown in (4)

$$\begin{cases} g^h = \sigma\left(F_h\left(f_m^h\right)\right) \\ g^w = \sigma\left(F_w\left(f_m^w\right)\right) \end{cases} \tag{4}$$

where F_h and F_w represent the convolution operation, and the kernel size is $C \times C/r \times 1 \times 1$. f_m^h and f_m^w represent MFH and MFW, respectively. σ denotes the sigmoid operation.

Subsequently, g^h and g^w are applied to the feature FM. By weighted summation of g^h, g^w, and FM, the attention to informative spectral in the precise positional region is achieved. The output of the coordinate attention feature in the cth channel is calculated as follows:

$$f_{e_c}^{co}(h,w) = f_c(h,w) \times g_c^h(h) \times g_c^w(w) \tag{5}$$

where $f_c(h,w)$ is the feature of FM in the cth channel, $f_{e_c}^{co}(h,w)$ is the feature in the cth channel, and superscript co represents that it is the coordinate feature.

To capture the spectral attention features with multiscale information, the multiscale features MF1, MF2, and MF3 with sizes $C/r \times 3 \times 3$, $C/r \times 6 \times 6$, and $C/r \times 8 \times 8$ are first convolved with convolution kernels of sizes $C \times C/r \times 3 \times 3$, $C \times C/r \times 6 \times 6$, and $C \times C/r \times 8 \times 8$, respectively. The spectral attention features $g^1 \in \mathbf{R}^{C\times1\times1}$, $g^2 \in \mathbf{R}^{C\times1\times1}$, and $g^3 \in \mathbf{R}^{C\times1\times1}$ at different scales are automatically learned, and the calculation process is shown in (6)

$$\begin{cases} g^1 = F_1\left(f_m^1\right) \\ g^2 = F_2\left(f_m^2\right) \\ g^3 = F_3\left(f_m^3\right) \end{cases} \tag{6}$$

where F_1, F_2, and F_3 represent the convolution operations with convolution kernel sizes of $C \times C/r \times 3 \times 3$, $C \times C/r \times 6 \times 6$, and $C \times C/r \times 8 \times 8$, respectively. f_m^1, f_m^2, and f_m^3 represent MF1, MF2, and MF3. g^1, g^2, and g^3 are then summed to generate $g^s \in \mathbf{R}^{C\times1\times1}$, which is used to weight the feature map FM. The expression of the multiscale output feature in the cth channel is formulated as

$$f_{e_c}^{sp}(h,w) = f_c(h,w) \times g_c^s(h,w) \tag{7}$$

where $f_{e_c}^{sp}(h,w)$ is the feature in the cth channel, and the superscript sp represents that it is the multiscale feature.

Finally, the feature map FM is combined with the spectral attention features through the skip connection to generate the spatial pyramid coordinate attention features. The output features in the cth channel are shown in (8)

$$f_{e_c}(h,w) = f_c(h,w) + f_{e_c}^{co}(h,w) + f_{e_c}^{sp}(h,w). \tag{8}$$

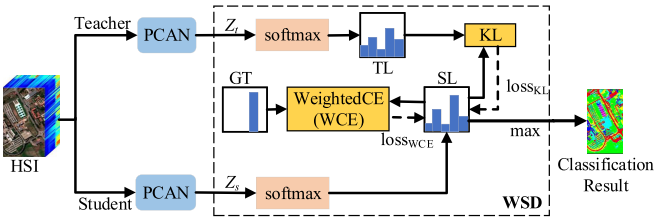


Fig. 4. Structure of WSD.

C. Weighted Self-Distillation

Existing DL algorithms usually take the difference between the prediction and the ground truth as the loss function. The prediction can gradually fit the ground truth by gradually reducing the loss. However, the ground truth adopts a one-hot encoding strategy: all negative labels are 0 except positive ones. This encoding strategy can only indicate which category an object belongs to and ignores the object-related information contained in other labels. When an object's characteristics are very close to several categories in real life, it is difficult for humans to identify which category it belongs to. At this point, it is crucial to use the remaining negative labels to reflect the similarity between objects and different categories. To further explore the prior information in the original image, this article first introduces the self-distillation method, which uses soft labels and ground truth as supervision information to guide students to train jointly. At the same time, considering the sample imbalance of HSI, a weighting strategy is introduced, and WSD is proposed to make the model pay more attention to small and indistinguishable samples. The structure of WSD is shown in Fig. 4.

As shown in Fig. 4, the teacher and student branches adopt the same model, both of which are PCAN.

Then the final classification result map is generated through WSD. Instead of training with one model alone, the self-distillation network uses soft label TL and ground-truth GT to guide student model training. The predicted label SL is generated by self-distillation, which further utilizes the prior information of the image. Specifically, the KL divergence is used to fit SL and TL, so that the student model can learn the parameter distribution of the teacher model. In addition, in the distillation process, the weight of the teacher model is fixed, and the generated supervision information is more accurate and stable, which helps students to explore the effective information in HSI further and accelerate the model convergence. Meanwhile, considering the imbalance of HSI samples, this article introduces weighted cross-entropy into WSD to fit SL and GT. Finally, the student network parameters are updated according to the backpropagation of the losses lossKL and lossWCE.

Specifically, the HSI is first fed into the teacher model PCAN and then softmax to generate the TL. Subsequently, fix the weights of the teacher model and feed the HSI into the student model PCAN to generate SL by softmax. The formulation of softmax is shown in (9)

$$p(z_i, T) = \frac{\exp(z_i/T)}{\sum_j \exp\left(z_j/T\right)} \tag{9}$$

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where z_i represents the logits of class i, and T is the temperature factor that controls the soft label contribution. When $T = 1$, it is the standard softmax function. $p(z_i, T)$ denotes the output probability of class i at temperature T. By introducing temperature, the proportion of negative labels is increased.

Classical cross-entropy usually performs poorly when dealing with imbalanced HSI samples. Therefore, WSD is proposed in this article. By adjusting the weight of each category loss, the similarity between ground truth and student prediction is evaluated. And more attention is paid to samples with small and indistinguishable samples. The weighted cross-entropy can be described as

$$L_{\mathrm{WCE}}(y, p(Z_s)) = -\sum_{i=1}^{n} w_i y \log(p(Z_s)) \tag{10}$$

where w_i is the weight of each category loss. Z_s is the logits of the student model, and $Z_s = [z_{s1}, z_{s2}, \ldots, z_{sC}]$, where C is the number of categories. The w_i is calculated in (11)

$$w_i = \frac{M}{F_i} \tag{11}$$

where M is the median frequency of all classes. F_i is the frequency of class i, which is defined as the ratio of the number of class i to the sum of nonbackground pixels in the training set. F_i can be defined as

$$F_i = \frac{N_i}{\sum_j N_j} \tag{12}$$

where N_i is the number of pixels of class i.

Then, the weighted cross-entropy L_{WCE} and KL divergence L_{KL} are used to update the student network by backpropagation. The total loss function can be described as

$$\begin{aligned} L_{\mathrm{WSD}} = {} & (1-\alpha) \times L_{\mathrm{WCE}}(y, p(Z_s)) \\ & + \alpha \times L_{\mathrm{KL}}(p(Z_t, T), p(Z_s, T)) \end{aligned} \tag{13}$$

where Z_s represents the logits of the student model and Z_t represents the logits of the teacher model. y is the ground truth and α is the loss balance coefficient to adjust the importance of L_{WCE} and L_{KL}.

D. Computational Complexity

As mentioned above, the PCAN mainly consists of three stages, and SPCS and SPCE generate each stage's computational complexity. SPCS mainly combines pyramid pooling and coordinate pooling to aggregate diverse spatial features. Assume that the size of the input image is $H \times W \times C$. After SPCS, features with spatial dimension N_s are generated, where $N_s = H + W + S$ and S is the total number of multiscale features. As described in Section II-A, $S = 9 + 64 + 81 = 109$. This process produces little computation. Then, feed it into SPCE. The channel is reduced to r/C through the full connection layer, and the computational complexity is $N_s \times C \times C/r$. Subsequently, divide the features into the size of H, W, 9, 36, and 64 from the spatial dimension. Then the features are reshaped and passed through 1×1, 3×3, 6×6, and 8×8 convolutions to increase the channel to C. The computational complexity is $C \times C/r \times 5$. The total computational complexity of PCAN is $3 \times C \times C/r \times (H + W + 114)$. On the contrary, the computational complexity of the original nonlocal model is $(H \times W) \times (H \times W) \times C$, and the computational complexity of ENL-FCN is $(H + W - 1) \times H \times W \times C$. It is roughly estimated that the computational complexity of ENL-FCN is $(H \times W \times r)/(3 \times C)$ times that of PCA-WSD.

Algorithm 1 PCAWSD

Input: HSI image X, ground truth L, the number of HSI categories C, training rate r, the maximum number of iterations N, and the phase of network P.

Output: Classification result map Y.

Initialization. Set the number of iterations n=0, the index set of all sample labels as I_{all}, the zero matrices L_{train} and L_{test} with the same shape as L as the training label and the testing label, the index set of train label $I_{train} \leftarrow \{\}$, the index set of test label $I_{test} \leftarrow \{\}$, and the index set of background label $I_{bg} \leftarrow \{\}$;
for *c=0 in C* **do**
  $I_c \leftarrow \{\}$ the position index of $L = c$;
  if $c = 0$ **then**
    $I_{bg} \leftarrow I_{bg} \cup I_c$;
  else
    The sample index i with size of $len(I_c) \times r$ is randomly selected from I_c;
    $I_{train} \leftarrow I_{train} \cup i$;
  end
end
$I_{test} \leftarrow I_{all} - I_{train}$;
$L_{train}(I_{train}) \leftarrow L(I_{train})$, $L_{test}(I_{test}) \leftarrow L(I_{test})$;
if $P ==' train'$ **then**
  Teacher branch:
  while $n < N$ **do**
    Feed X and I_{train} into the teacher model;
    Perform SPCS by (1) and (2) to aggregate diverse spatial features;
    Perform SPCE to the aggregated features, by (3)-(8);
    Generate the output Z_t of the teacher model;
  end
  Student branch:
  while $n < N$ **do**
    Feed X and L_{train} into the student model;
    Repeat step 17-19 to generate the output Z_s of the student model;
    Calculate the loss of WSD through Z_t and Z_s by (9)-(13);
  end
else
  Feed X into the student model to generate the Y;
end
Return Y.

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TABLE I

SAMPLE NUMBER IN EACH CATEGORY FOR THE TRAINING SET AND TESTING SET ON IP

No.	Color	Category name	Sample	Training	Test
0		Background	10776		
1		Alfalfa	46	5	41
2		Corn-notill	1428	143	1285
3		Corn-mintill	830	83	747
4		Corn	237	24	213
5		Grass/Pasture	483	49	434
6		Grass/Trees	730	73	657
7		Grass/Pasture-mowed	28	3	25
8		Hay-windrowed	478	48	430
9		Oats	20	2	18
10		Soybean-notill	972	98	874
11		Soybean-mintill	2455	246	2209
12		Soybean-clean	593	60	533
13		Wheat	205	21	184
14		Woods	1265	127	1138
15		Buildings-Grass-Trees-Drivers	386	39	347
16		Stone-Steel Towers	93	10	83
		Total	10249	1031	9218

This article proposes the PCA-WSD algorithm, and the basic steps are shown in Algorithm 1.

III. EXPERIMENT AND ANALYSIS

To evaluate the effectiveness of the proposed PCA-WSD algorithm, we conduct experiments on four public benchmark datasets.

A. Dataset Description

The four public HSI datasets include Indian Pine (IP), Pavia University (UP), Kennedy Space Center (KSC), and Pavia Center (PC). They are public and can be accessed online.

1) IP Dataset: The IP dataset contains 200 bands and 145 × 145 pixels with a spatial resolution of 20 m/pixel. There are 10 249 labeled pixels, covering 16 types of wetland land cover. The 10% samples of each class are randomly selected as training sets, and the remaining as testing sets. Table I lists the details of the IP dataset, including the category name, the number of samples, training samples, and testing samples per category.

2) UP Dataset: The UP dataset contains 103 bands and 610 × 340 pixels with a spatial resolution of 1.3 m/pixel. There are 42 776 labeled pixels, covering nine types of land cover. The 1% samples of each class are randomly selected as training sets and the remaining as testing sets. Table II lists the details of the UP dataset, including category name, the number of samples, training samples, and testing samples per category.

3) KSC Dataset: The KSC dataset contains 176 bands and 512 × 614 pixels with a spatial resolution of 18 m/pixel. There are 5211 labeled pixels, covering 13 types of land cover. The 5% samples of each class are randomly selected as training sets and the remaining as testing sets. Table III lists the details of the KSC dataset, including category name, the number of samples, training samples, and testing samples per category.

TABLE II

SAMPLE NUMBER IN EACH CATEGORY FOR THE TRAINING SET AND TESTING SET ON UP

No.	Color	Category name	Sample	Training	Test
0		Background	164624		
1		Asphalt	6631	67	6534
2		Meadows	18649	187	17897
3		Gravel	2099	21	2078
4		Trees	3064	31	3033
5		Painted metal sheets	1345	14	1331
6		Bare Soil	5029	51	4978
7		Bitumen	1330	14	1316
8		Self-Blocking Bricks	3682	37	3645
9		Shadows	947	10	937
		Total	42776	432	42344

TABLE III

SAMPLE NUMBER IN EACH CATEGORY FOR THE TRAINING SET AND TESTING SET ON KSC

No.	Color	Category name	Sample	Training	Test
0		Background	309157		
1		Scrub	761	39	722
2		Willow swamp	243	13	230
3		Cabbage palm hammock	256	13	243
4		Slash pine	252	13	239
5		Oak/Broadleaf hammock	161	9	152
6		Hardwood	229	12	217
7		Swamp	105	6	99
8		Graminoid-marsh	431	22	409
9		Spartina-marsh	520	16	494
10		Cattail marsh	404	21	383
11		Salt marsh	419	21	398
12		Mud flats	503	26	477
13		Water	927	47	880
		Total	5211	268	4943

TABLE IV

SAMPLE NUMBER IN EACH CATEGORY FOR THE TRAINING SET AND TESTING SET ON PC

No.	Color	Category name	Sample	Training	Test
0		Background	635488		
1		Water	65971	660	65311
2		Trees	7598	76	7522
3		Asphalt	3090	31	3059
4		Self-Blocking Bricks	2685	27	2658
5		Bitumenh	6584	66	6518
6		Tiles	9248	93	9155
7		Shadows	7287	73	7214
8		Meadows	42826	429	42397
9		Bare Soil	2863	29	2834
		Total	783640	1484	146668

4) PC Dataset: The PC dataset contains 102 bands and 1096 × 715 pixels with a spatial resolution of 1.3 m/pixel. There are 148 152 labeled pixels, covering nine types of land cover. The 1% samples of each class are randomly selected as training sets and the remaining as testing sets. Table IV lists the details of the PC dataset, including category name,

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TABLE V
CLASSIFICATION RESULTS OF EIGHT METHODS ON CA, OA, AA, AND KAPPA OF THE IP DATASET

Class	Contextual	SSRN	DBDA	HPDM-SPRN	A^2S^2K-ResNet	MSRN	ENL-FCN	PCA-WSD
1	75.82±10.43	74.22±38.41	96.45±5.18	95.12±7.71	98.48±2.33	98.57±2.86	98.29±1.12	**100.00±0.00**
2	96.87±1.45	**98.61±0.65**	98.19±0.77	98.24±0.84	98.29±1.09	95.04±7.47	96.97±0.65	98.04±0.46
3	95.40±2.23	98.15±0.32	97.50±2.54	97.12±10.42	98.62±0.86	98.75±1.05	99.45±0.37	**99.99±0.04**
4	93.29±2.57	98.76±1.15	98.10±2.26	97.93±1.66	98.59±1.68	99.06±1.17	96.90±0.87	**99.20±0.70**
5	96.13±1.70	94.75±2.51	97.32±2.40	95.61±2.44	98.94±0.93	99.33±0.49	**99.12±0.56**	98.92±0.44
6	99.28±0.43	98.73±0.66	98.30±1.53	**99.88±0.11**	98.96±1.02	99.16±0.63	99.01±0.51	99.74±0.10
7	83.62±17.49	58.46±0.48	90.66±15.89	98.00±4.10	98.02±4.53	94.38±11.25	98.00±2.67	**100.00±0.00**
8	96.21±1.60	96.21±1.60	**100.00±0.00**	99.98±0.07	99.49±0.95	99.49±1.01	99.60±0.30	99.98±0.07
9	85.45±13.73	60.00±48.99	86.02±14.76	93.89±15.00	86.82±17.11	81.03±10.08	81.11±8.31	**99.44±1.67**
10	97.48±1.16	98.35±0.45	96.19±1.42	96.45±1.61	97.44±1.17	98.76±0.92	98.30±0.50	**99.13±0.41**
11	97.88±0.64	98.95±0.89	99.10±0.47	98.56±0.90	99.09±0.48	99.28±0.35	99.77±0.04	**99.69±0.10**
12	97.14±1.08	97.25±2.10	98.34±1.11	98.99±0.44	97.78±1.92	99.14±0.51	99.00±0.41	**99.25±0.21**
13	93.65±5.51	98.17±3.10	99.21±1.23	**99.21±1.23**	99.18±1.81	99.28±0.95	98.58±0.70	99.29±0.42
14	97.46±2.15	99.32±0.59	99.47±0.51	99.68±0.27	99.11±0.71	99.19±0.92	99.91±0.08	**99.92±0.10**
15	93.68±3.53	97.84±2.78	98.20±1.28	95.01±5.04	99.13±0.74	**99.23±0.53**	96.08±2.10	98.67±0.61
16	92.18±7.43	83.98±12.43	94.08±5.93	98.21±3.54	95.51±3.84	95.08±5.53	93.98±2.75	**99.76±0.48**
OA(%)	96.79±0.61	98.18±0.56	98.20±0.30	98.23±0.39	98.63±0.48	98.35±1.52	98.77±0.18	**99.36±0.09**
AA(%)	93.22±2.07	90.83±7.12	96.31±1.38	97.64±1.29	97.72±1.18	97.17±1.45	97.13±0.63	**99.44±0.15**
Kappa(×100)	96.33±0.69	97.93±0.64	97.95±0.34	97.98±0.44	98.44±0.55	98.12±1.73	98.59±0.21	**99.27±0.11**

the number of samples, training samples, and testing samples per category.

B. Experimental Setting

This article evaluates the classification performance from both qualitative and quantitative perspectives. For qualitative analysis, the classification map is used to visualize the classification result. For quantitative analysis, four metrics are adopted to achieve a more accurate analysis, including category accuracy (CA), overall accuracy (OA), average accuracy (AA), and Kappa coefficient. Here, CA is the classification accuracy of each category of samples, OA is the ratio of the number of correctly predicted samples to that of all test samples, AA is the average of the CA of all categories, and Kappa coefficient measures the consistency between the classification result and the ground truth. The larger the value of these four metrics, the better the classification result. Moreover, all experiments are repeated ten times to avoid estimation bias, and the final results are recorded as "mean accuracy ± standard deviation."

During training, the number of epochs and batch size are set to 600 and 1, respectively. The Adam optimizer is used to optimize the network, where weight decay and the learning rate are set to 0.01 and 8e-4. The temperature parameters T and α in WSD are set to 4 and 0.01. All experiments are implemented in PyTorch environment with Intel Xeon CPU E5-2630 v4@2.20 GHz, 64 GB RAM, NVIDIA GeForce GTX TITAN Xp graphics processing unit (GPU), 12 GB RAM, based on Ubuntu 18.04 Ultimate 64-bit system.

In this article, we take the whole image $X \in R^{H\times W\times C}$ as the input and use a masking strategy to select the training and testing sets, which are the same size as the whole image and have labels only at specific indexes. Specifically, the indexes of nonbackground pixels are first obtained. Then a certain percentage of indexes are randomly selected according to the category to add to the training set indexes, and the rest to add to the testing set indexes. In addition, the image is normalized to standard normal distribution before being fed into the network.

C. Experimental Results and Analysis

In this article, the latest seven algorithms are selected for comparison. According to the format of the input data, these comparative algorithms can be divided into patch- and whole-image-based HSI classification methods. Among them, Contextual [43], SSRN [21], double-branch dual-attention mechanism network (DBDA) [44], HPDM-SPRN [30], A^2S^2K-ResNet [45], and MSRN [7] are patch-based classification methods, while ENL-FCN [36] is the whole-image-based classification method.

1) Experimental Results and Analysis of the IP Dataset: For the IP dataset, Table V shows the classification results of PCA-WSD and its comparative algorithms, where the best results are shown in bold.

As shown in Table V, the proposed PCA-WSD algorithm performs best in OA, AA, and Kappa, reaching 99.36%, 99.44%, and 99.27%, respectively. Compared with the suboptimal results, they increased by 0.59%, 1.8%, and 0.68%. The PCA-WSD improves AA more significantly, which is attributed mainly to WSD. It is well known that the IP dataset is highly imbalanced. In some categories, the samples' number can only reach single digit, for example, Alfalfa, Grass/Pasture mowed, Oats, and Stone-Steel Towers, which contain five, three, two, and ten samples. The designed WSD makes the model pay more attention to small and indistinguishable samples by assigning loss weights to different categories. The CA of PCA-WSD in these categories reaches 100.00%, 100.00%, 99.44%, and 99.76%, which is 1.71%, 2.00%, 5.55%, and 1.55% higher than that of suboptimal, respectively. Experiments fully demonstrate the efficiency of WSD in solving the imbalance of samples.

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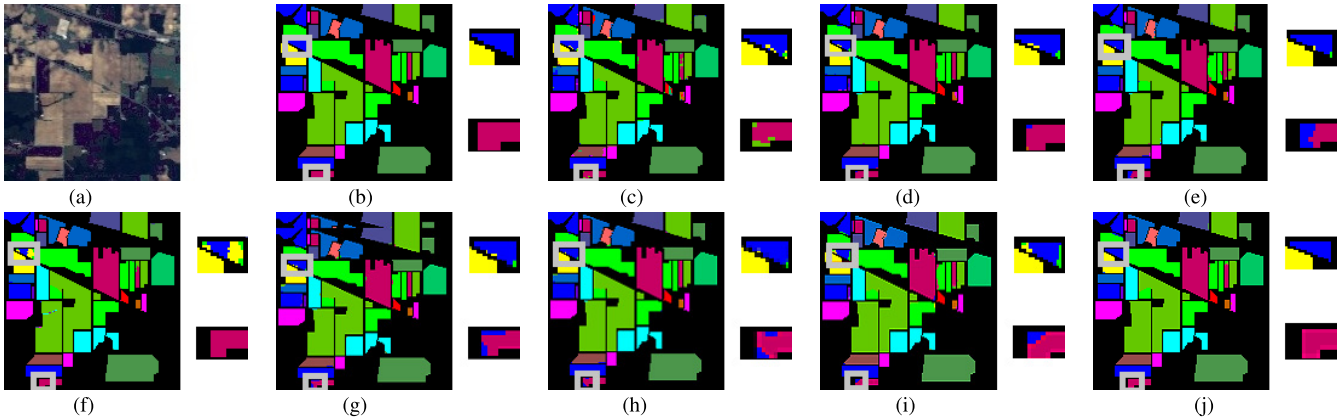


Fig. 5. Classification maps of different methods on the IP dataset. (a) False-color image. (b) Ground truth. (c) Contextual. (d) SSRN. (e) DBDA. (f) HPDM-SPRN. (g) A^2S^2K-ResNet. (h) MSRN. (i) ENL-FCN. (j) PCA-WSD.

TABLE VI
CLASSIFICATION RESULTS OF EIGHT METHODS ON CA, OA, AA, AND KAPPA OF THE UP DATASET

Class	Contextual	SSRN	DBDA	HPDM-SPRN	A^2S^2K-ResNet	MSRN	ENL-FCN	PCA-WSD
1	77.39±6.96	88.53±3.67	94.90±1.25	98.97±0.76	93.53±2.50	98.17±0.30	**99.17±0.36**	99.05±0.43
2	93.45±1.44	95.95±4.41	98.59±0.49	99.88±0.13	98.56±0.34	99.16±0.11	99.78±0.23	**99.97±0.03**
3	57.03±10.58	78.08±16.04	89.11±5.98	86.14±8.17	86.70±10.20	87.69±10.78	89.69±2.11	**98.15±0.70**
4	83.67±9.03	**99.52±0.42**	98.93±0.76	96.11±1.83	98.98±1.31	99.83±0.14	97.55±0.68	98.78±0.24
5	96.07±7.14	99.73±0.39	98.87±0.97	99.90±0.19	99.40±0.86	99.77±0.16	99.69±0.81	**100.00±0.00**
6	92.13±5.06	97.24±1.76	98.85±0.65	97.74±1.53	97.92±1.76	99.02±0.84	99.59±0.36	**99.98±0.04**
7	69.12±1.98	85.86±18.04	90.91±6.15	98.78±2.04	96.31±2.43	99.27±1.06	98.31±1.26	**99.96±0.11**
8	56.61±9.25	85.82±5.51	84.98±3.13	95.84±1.63	85.98±2.00	88.25±1.84	98.64±0.47	**99.62±0.23**
9	78.26±11.67	78.37±39.22	97.23±2.68	96.78±3.71	98.53±1.71	97.82±1.77	97.85±0.93	**99.48±0.36**
OA(%)	84.55±2.18	92.43±2.36	96.06±0.49	98.11±0.38	95.90±0.96	97.39±0.72	98.82±0.21	**99.62±0.09**
AA(%)	78.19±3.34	89.90±6.11	94.71±0.84	96.71±0.92	95.10±1.52	96.56±1.42	97.81±0.32	**99.44±0.13**
Kappa(×100)	79.20±2.97	89.85±3.23	94.76±0.65	97.49±0.50	94.56±1.27	96.54±0.95	98.43±0.27	**99.49±0.12**

The classification result maps of these algorithms on the IP dataset are shown in Fig. 5. In Fig. 5(c) and (d), Context and SSRN algorithms show significant noise in the wheat, soybean-notill, and buildings grass–trees–drivers categories, likely due to ignoring the capture of long-distance context. In Fig. 5(e)–(i), although all these methods use the attention mechanism, due to the lack of joint attention on spectral–spatial features, lots of noise appears at the feature boundary, resulting in boundary ambiguity. Especially in places with few samples, such as grass/pasture-mowed, oats, and stone-steel towers, there are many misclassifications in the comparative algorithms. As can be seen from Fig. 5(j), the classification results of the PCA-WSD algorithm are smooth at the boundary and uniform inside the image, especially for small samples PCA-WSD achieves considerable results.

2) Experimental Results and Analysis of the UP Dataset: For the UP dataset, the classification results evaluated on PCA-WSD and its comparative algorithms are shown in Table VI, where the best results are shown in bold.

As shown in Table VI, compared with other advanced algorithms, the proposed PCA-WSD achieves the best performance in OA, AA, and Kappa, reaching 99.62%, 99.44%, and 99.49%, respectively. PCA-WSD achieves the best CA in seven of the nine categories of samples, while the remaining two categories are optimal when using ENL-FCN and SSRN algorithms, respectively. The Gravel, Bitumen, and Shadows have 21, 14, and 10 samples. The CA of PCA-WSD on them is 98.15%, 99.96%, and 99.48%, which is 8.46%, 0.69%, and 0.95% higher than the suboptimal ones. Experimental results show that PCA-WSD can effectively extract beneficial features for classification. In addition, it can be seen from the variance of CA that PCA-WSD has good robustness. In 10 Monte Carlo operations, the fluctuation range of experimental results is small. Especially for Gravel, the variance of other algorithms is more than 2%, while PCA-WSD is only 0.7%.

For the UP dataset, the classification result maps of PCA-WSD and its comparative algorithms are shown in Fig. 6. In Fig. 6(c)–(e), the experimental results of Contextual, SSRN, and DBDA have a lot of noise in homogeneous areas (such as bare soil) and banded areas (such as self-blocking bricks and gravel), which cannot handle local and long-range features well. Compared with the above three algorithms, the algorithm in Fig. 6(f)–(i) can obtain better results, but still has the problem of misclassification in the case of fewer samples. As shown in Fig. 6(j), the proposed PCA-WSD has smoother classification results and can capture local context

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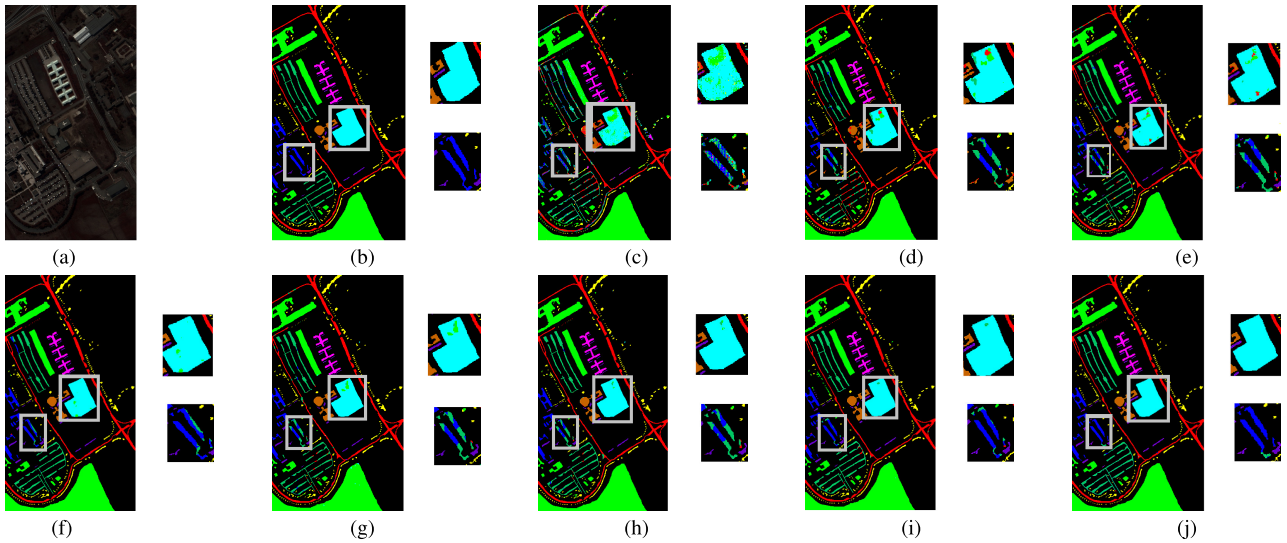


Fig. 6. Classification maps of different methods on the UP dataset. (a) False-color image. (b) Ground truth. (c) Contextual. (d) SSRN. (e) DBDA. (f) HPDM-SPRN. (g) A^2S^2K-ResNet. (h) MSRN. (i) ENL-FCN. (j) PCA-WSD.

TABLE VII
CLASSIFICATION RESULTS OF EIGHT METHODS ON CA, OA, AA, AND KAPPA OF THE KSC DATASET

Class	Contextual	SSRN	DBDA	HPDM-SPRN	A^2S^2K-ResNet	MSRN	ENL-FCN	PCA-WSD
1	98.08±1.23	99.65±0.51	99.94±0.12	99.76±0.32	99.43±0.99	**100.00±0.00**	98.64±0.66	99.70±0.27
2	94.24±5.93	96.79±3.68	98.85±1.87	97.66±2.30	96.65±5.18	99.14±1.72	99.96±0.13	**100.00±0.00**
3	62.31±7.07	98.51±2.98	97.88±4.24	97.16±2.57	96.93±3.66	97.90±3.77	99.42±0.38	**99.79±0.21**
4	69.48±11.26	99.15±0.97	98.00±1.65	74.56±9.81	99.26±1.05	99.13±0.81	99.71±0.33	**100.00±0.00**
5	71.28±8.56	93.18±11.36	87.84±9.68	60.78±11.98	90.65±10.51	92.48±7.31	99.80±0.42	**100.00±0.00**
6	93.05±8.03	99.32±0.65	99.90±0.19	89.08±8.71	99.43±0.75	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
7	73.48±13.99	97.07±1.66	**100.00±0.00**	59.10±36.52	99.59±0.82	99.78±0.44	99.90±0.30	99.80±0.40
8	98.64±1.53	98.49±1.41	98.65±1.24	98.12±2.48	99.24±0.91	99.49±0.23	99.51±0.54	**99.98±0.07**
9	98.27±1.46	99.21±1.17	**100.00±0.00**	97.49±5.18	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
10	97.98±1.44	**100.00±0.00**	99.84±0.33	98.96±1.33	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
11	93.14±13.33	98.82±2.36	99.59±0.83	99.82±0.30	99.95±0.11	97.76±1.28	**100.00±0.00**	**100.00±0.00**
12	97.58±2.18	94.31±9.67	99.04±.0.85	94.83±5.72	**99.82±0.36**	99.82±0.35	96.98±0.68	96.33±0.21
13	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
OA(%)	92.37±2.86	98.31±2.24	99.01±0.50	94.97±0.62	99.05±0.48	99.28±0.44	99.42±0.09	**99.59±0.02**
AA(%)	88.27±2.68	98.04±2.22	98.43±0.89	89.90±2.08	98.53±0.84	98.88±0.75	99.53±0.10	**99.66±0.03**
Kappa(×100)	91.50±3.18	98.12±2.49	98.90±0.56	94.40±0.69	98.94±0.54	99.20±0.49	99.35±0.10	**99.54±0.03**

and long-range dependencies well. Especially for the small samples, it achieves significant results.

3) Experimental Results and Analysis of the KSC Dataset: For the KSC dataset, the classification results evaluated on PCA-WSD and its comparative algorithms are shown in Table VII, with the best results in bold.

As shown in Table VII, the proposed PCA-WSD algorithm achieves the best performance in three metrics compared to other advanced algorithms. Specifically, the OA, AA, and Kappa can reach 99.59%, 99.66%, and 99.54%, respectively, which are 0.17%, 0.13%, and 0.19% higher than the sub-optimal results. Compared with the IP and UP datasets, the PCA-WSD algorithm has less significant improvement on the KSC dataset, mainly because the KSC dataset is relatively balanced. In addition, PCA-WSD achieves the best CA in 10 of the 13 categories, of which CA reaches 100% in eight categories because PCA-WSD employs the PCAM that can extract more discriminative features. Meanwhile, the mean and covariance of the metrics were obtained through 10 Monte Carlo experiments. It can be found that, compared with other algorithms, the experimental results of the PCA-WSD algorithm are more stable and robust.

This article visualizes the classification maps of PCA-WSD and other comparison algorithms on the KSC dataset. The visualized classification maps are shown in Fig. 7. It can be seen that compared with IP and UP datasets, the results of these methods on the KSC dataset are more stable and less noisy. The error-prone classification points are mainly concentrated in the gray box area in Fig. 7. By zooming in on the gray box, it can be seen that the gray boxes mainly

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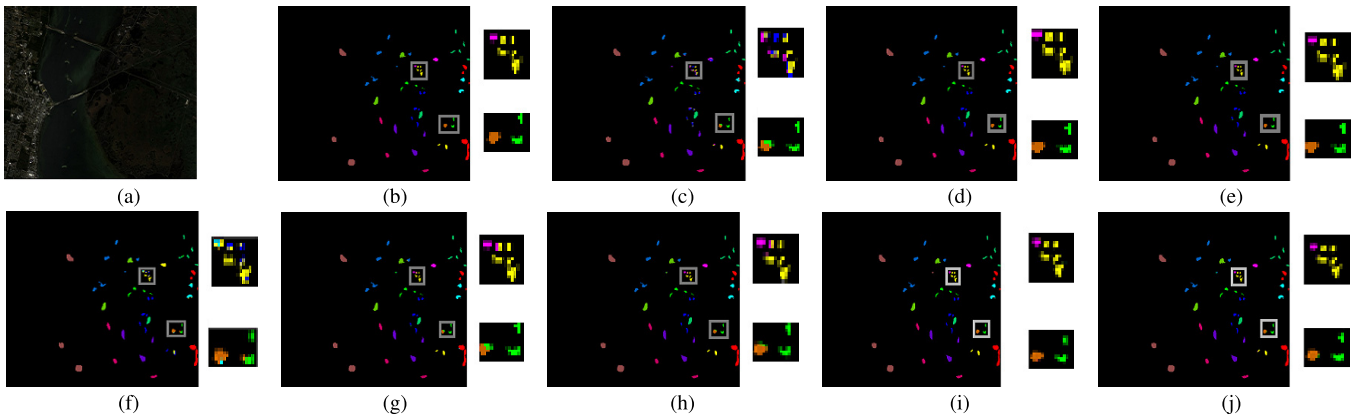


Fig. 7. Classification maps of different methods on the KSC dataset. (a) False-color image. (b) Ground truth. (c) Contextual. (d) SSRN. (e) DBDA. (f) HPDM-SPRN. (g) A^2S^2K-ResNet. (h) MSRN. (i) ENL-FCN. (j) PCA-WSD.

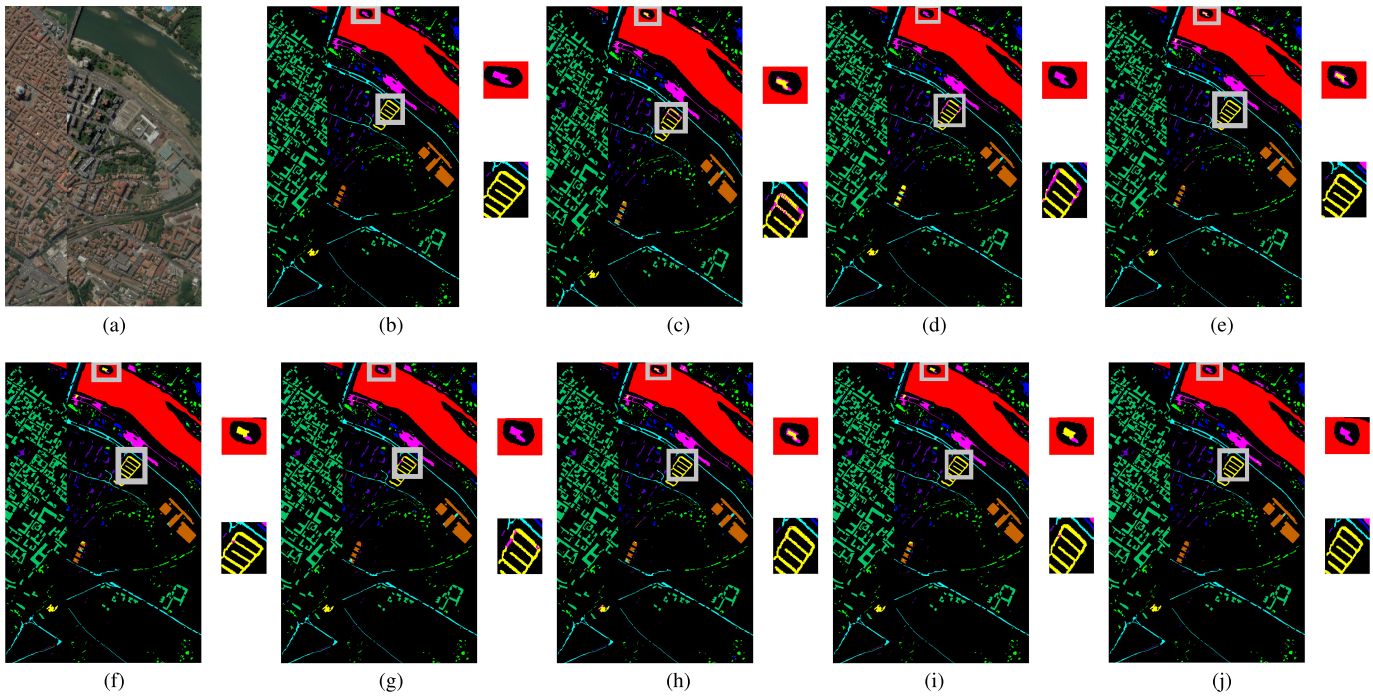


Fig. 8. Classification maps of different methods on the PC dataset. (a) False-color image. (b) Ground truth. (c) Contextual. (d) SSRN. (e) DBDA. (f) HPDM-SPRN. (g) A^2S^2K-ResNet. (h) MSRN. (i) ENL-FCN. (j) PCA-WSD.

contain slash pine, oak/broadleaf hammock, swamp, and willow swamp land covers. Referring to Table III, we know that the number of training sets in these categories is 13, 9, 6, and 13, respectively. As seen from Fig. 7, the Contextual and HPDM-SPRN perform poorly in classifying slash pine, oak/broadleaf hammock, and swamp. Contextual, SSRN, and A^2S^2K-ResNet have poor classification effects on the willow swamp. The proposed PCA-WSD algorithm performs well in these categories with small samples. This phenomenon is consistent with the results in Table VII.

4) Experimental Results and Analysis of the PC Dataset: For the PC dataset, the classification results evaluated on PCA-WSD and its comparative algorithms are shown in Table VIII, where the best results are shown in bold.

As shown in Table VIII, compared with other advanced algorithms, the proposed PCA-WSD algorithm achieves the best performance in OA, AA, and Kappa, reaching 99.71%, 99.33%, and 99.59%, respectively. Compared with the suboptimal results, they increased by 0.34%, 1.27%, and 0.48%, respectively. PCA-WSD achieves the best CA in six of the nine categories of samples, while the remaining three categories are optimal when using ENL-FCN and Contextual algorithms. Especially for asphalt, self-blocking bricks, and bare soil with 31, 27, and 29 training samples, the CA of PCA-WSD is 99.57%, 99.90%, and 99.76%, respectively. The experimental results show that the PCA-WSD can extract more discriminative features from small data conducive to classification, which alleviating the problem of sample imbalance effectively.

The classification maps of the PCA-WSD algorithm and its comparative algorithms on the PC are shown in Fig. 8. In 8(c)–(e), the experimental results of Contextual, SSRN, and DBDA have a lot of noise in homogeneous areas (such

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TABLE VIII
CLASSIFICATION RESULTS OF EIGHT METHODS ON CA, OA, AA, AND KAPPA OF THE PC DATASET

Class	Contextual	SSRN	DBDA	HPDM-SPRN	A^2S^2K-ResNet	MSRN	ENL-FCN	PCA-WSD
1	99.90±0.08	99.97±0.04	99.98±0.02	**100.00±0.00**	99.99±0.00	99.87±0.12	**100.00±0.00**	**100.00±0.00**
2	96.56±1.58	94.52±5.28	97.50±0.99	96.06±1.89	96.62±1.61	94.58±0.89	**97.57±1.61**	96.50±0.22
3	86.42±2.79	83.93±6.47	84.43±4.75	94.02±3.42	84.32±3.93	93.89±1.83	93.43±4.86	**99.57±0.26**
4	85.16±2.73	88.18±9.66	92.74±3.18	97.39±1.88	91.32±7.28	86.18±2.54	92.90±4.94	**99.90±0.06**
5	93.22±2.60	91.89±4.08	96.02±2.79	98.47±1.64	96.25±2.40	98.83±0.76	91.64±8.05	**99.67±0.21**
6	96.18±1.16	97.50±1.21	96.64±1.11	99.31±0.40	97.17±0.61	98.65±0.14	99.58±0.34	**99.66±0.19**
7	99.59±0.41	**99.75±0.29**	99.58±0.56	97.75±0.82	99.62±0.49	94.72±1.15	96.36±1.81	99.00±0.48
8	99.71±0.11	99.95±0.04	99.76±0.14	99.91±0.10	99.96±0.01	99.80±0.05	**99.98±0.00**	99.96±0.02
9	98.37±0.92	99.31±0.57	99.45±0.42	99.63±0.31	99.76±0.19	94.00±4.27	98.65±1.76	**99.76±0.21**
OA(%)	98.53±0.19	98.50±0.32	98.86±0.18	99.37±0.09	98.91±0.26	98.72±0.07	99.00±0.46	**99.71±0.04**
AA(%)	95.01±0.43	95.00±1.01	96.23±0.68	98.06±0.34	96.11±0.34	95.61±0.26	96.68±1.60	**99.33±0.09**
Kappa(×100)	97.91±0.27	97.87±0.45	98.39±0.26	99.11±0.13	98.46±0.37	98.18±0.09	98.58±0.65	**99.59±0.05**

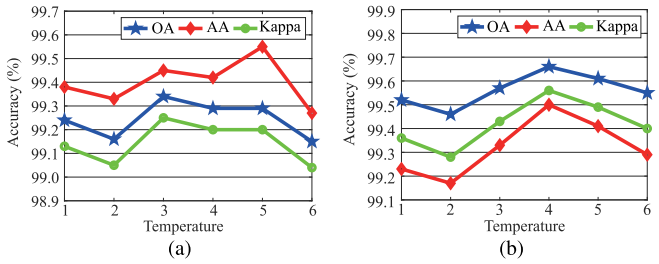


Fig. 9. Classification results of IP and UP datasets at different temperatures. (a) IP. (b) UP.

as bare soil) and banded areas (such as self-blocking bricks and gravel). They cannot extract multiscale local and long-range features effectively. Compared with the above three algorithms, the algorithm in Fig. 8(f)–(i) can obtain better classification performance. However, it still has the problem of misclassification when the number of samples is small. As shown in Fig. 8(j), the proposed PCA-WSD has smoother classification results and can capture local context and long-range dependencies well. Especially for the small samples, it achieves significant results.

D. Parameter Analysis

In the WSD, temperature T is a vital hyperparameter that affects the classification. The T is set to 1–6 to verify the effect of different temperatures. The experimental results on different datasets are shown in Fig. 9.

As shown in Fig. 9, the OA and AA of the two datasets generally show a zigzag trend. The IP and UP datasets first decrease, then increase, and then decrease with the increase of T. For the IP dataset, considerable results are obtained when $T = 3$, 4, and 5, among which the best result is obtained when $T = 3$. For the UP dataset, when $T = 3$, 4, and 5, considerable results are obtained, of which $T = 4$ is the best. In conclusion, when $T = 4$, considerable results can be achieved on these datasets. Therefore, this article chooses T to be 4.

E. Ablation Experiment

To further verify the effectiveness of PCA-WSD, ablation experiments are conducted from three aspects: the effectiveness of the component in PCA-WSD, the component in SPCS, and WSD.

TABLE IX
ABLATION EXPERIMENTAL RESULTS OF DIFFERENT COMPONENTS IN PCA-WSD ON THE IP DATASET

class	w/o PCAM	w/o WSD	PCA-WSD
1	100.00±0.00	96.59±1.19	100.00±0.00
2	97.62±0.37	98.30±0.29	98.04±0.46
3	99.91±0.10	99.97±0.05	99.99±0.04
4	97.46±0.94	97.28±0.75	99.20±0.70
5	99.22±0.43	99.91±0.18	98.92±0.44
6	99.73±0.16	99.21±0.35	99.74±0.10
7	100.00±0.00	98.40±4.80	100.00±0.00
8	99.95±0.14	100.00±0.00	99.98±0.07
9	98.33±3.56	27.78±16.85	99.44±1.67
10	97.55±0.56	98.76±0.16	99.13±0.41
11	99.59±0.18	99.84±0.06	99.69±0.10
12	99.14±0.30	99.40±0.37	99.25±0.21
13	99.67±0.43	98.59±0.55	99.29±0.42
14	99.96±0.07	99.92±0.03	99.92±0.10
15	93.63±1.24	97.81±1.40	98.67±0.61
16	99.88±0.36	97.60±1.20	99.76±0.48
OA(%)	98.91±0.11	99.14±0.12	**99.36±0.09**
AA(%)	98.85±0.30	93.54±0.66	**99.44±0.15**
Kappa(×100)	98.75±0.12	99.02±0.14	**99.27±0.11**

1) Effectiveness of Each Component of PCA-WSD: To verify the effectiveness of components in PCA-WSD, three sets of experiments are conducted, namely w/o PCAM, w/o WSD, and PCA-WSD on the IP dataset, as shown in Table IX.

Among them, w/o PCAM means PCA-WSD without PCAM, and w/o WSD means PCA-WSD without WSD. As shown in Table IX, PCA-WSD achieves the best experimental results of OA, AA, and Kappa on the IP dataset. Specifically, compared with w/o PCAM, PCA-WSD increases OA, AA, and Kappa from 98.91%, 98.85%, and 98.75% to 99.36%, 99.44%, and 99.27%, respectively. Compared with w/o WSD, the PCA-WSD algorithm has higher performance for AA, with an immediate increase of 5.90%. It matches the dataset distribution, where the IP dataset samples are relatively imbalanced. In particular, there are only two training samples for the ninth class of the IP dataset. After repeated

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TABLE X
ABLATION CLASSIFICATION RESULTS OF DIFFERENT COMPONENTS IN SPCS ON IP, UP, AND KSC DATASETS

dataset	method	OA	AA	Kappa
IP	w/o spp	99.35±0.09	99.39±0.17	99.26±0.10
	w/o coord	98.87±0.13	98.61±0.32	98.72±0.14
	w/o coord +global	98.91±0.10	98.80±0.19	98.75±0.12
	SPCS	**99.36±0.09**	**99.44±0.15**	**99.27±0.11**
UP	w/o spp	99.34±0.17	99.18±0.21	99.13±0.23
	w/o coord	99.36±0.12	99.02±0.13	99.15±0.16
	w/o coord +global	99.40±0.08	99.11±0.10	99.20±0.10
	PCA-WSD	**99.62±0.09**	**99.44±0.13**	**99.49±0.12**
KSC	w/o spp	99.48±0.72	99.54±0.11	99.43±0.08
	w/o coord	99.54±0.07	99.63±0.06	99.49±0.08
	w/o coord +global	99.58±0.07	99.65±0.07	99.53±0.08
	PCA-WSD	**99.59±0.02**	**99.66±0.03**	**99.54±0.03**

TABLE XI
ABLATION EXPERIMENTAL RESULTS OF WSD ON IP, UP, AND KSC DATASETS

dataset	method	weight	SD	OA	AA	Kappa
IP	w/o WSD			99.14±0.12	93.54±0.66	99.02±0.14
	w/o T	✓		99.33±0.06	99.27±0.25	99.23±0.07
	w/o Sw		✓	99.18±0.08	94.33±1.19	99.07±0.09
	WSD	✓	✓	**99.36±0.09**	**99.44±0.15**	**99.27±0.11**
UP	w/o WSD			99.39±0.11	98.90±0.23	99.20±0.14
	w/o T	✓		99.42±0.14	99.16±0.19	99.23±0.18
	w/o Sw		✓	99.56±0.12	99.28±0.23	99.41±0.16
	WSD	✓	✓	**99.62±0.09**	**99.44±0.13**	**99.49±0.12**
KSC	w/o WSD			99.50±0.09	99.45±0.17	99.44±0.10
	w/o T	✓		99.51±0.06	99.57±0.05	99.50±0.07
	w/o Sw		✓	99.37±0.17	99.28±0.33	99.30±0.19
	WSD	✓	✓	**99.59±0.02**	**99.66±0.03**	**99.54±0.03**

experiments, the average CA of the w/o WSD in this class is only 27.78%, while the PCA-WSD can reach 99.44%, which is the main reason for the big difference in AA. In summary, the large gap is that the IP dataset is relatively unbalanced. In addition, WSD mainly considers the problem of sample imbalance, which can improve the accuracy of samples with small sample sizes, thus improving the AA.

2) Effectiveness of Each Component in SPCS: To verify the effectiveness of the SPCS, this article ablates the components used to aggregate local and global features, respectively. There are four experiments: w/o spp, w/o coord, w/o coord + global, and SPCS. Among them, w/o spp refers to the SPCS after removing the component used to capture multiscale local information. w/o coord refers to the SPCS after removing the component used to capture positional information. And w/o coord + global refers to adding global max pooling based on w/o spp. The results are shown in Table X.

As shown in Table X, the OA, AA, and Kappa of SPCS achieve the best results on datasets. Experiments show that SPCS with both multiscale and positional information can effectively extract more discriminative features. The performance of w/o coord + global is slightly better than that of w/o coord, indicating that the global information can enrich the extracted features. Compared with the w/o coord + global, the performance improvement of SPCS is noticeable on IP and UP datasets. It indicates that the global features with positional information are still quite significant. While KSC has block distribution among various classes, it is essential to have multiscale local information. Therefore, compared with w/o coord, the performance of w/o spp degrades more.

3) Effectiveness of WSD: To verify the effectiveness of WSD, four sets of experiments are conducted on IP, UP, and KSC datasets, including w/o WSD, w/o T, w/o Sw, and WSD, as shown in Table XI. Among them, w represents that weighting is used to calculate the cross-entropy, and SD represents that self-distillation is used in the network. Specifically, w/o WSD means the network does not contain the WSD, and w/o T means there is no teacher network. The w/o Sw means the student network is not weighted in calculating cross-entropy. WSD is the algorithm proposed in this article, which includes the teacher network, and both the teacher network and the student network use weighted cross-entropy.

It can be seen from Table XI that w/o WSD obtains the worst results on the three datasets, especially for the AA on the IP and UP datasets, which can only reach 93.54% and 98.90%. In contrast, the proposed WSD algorithm achieves the best results on all datasets, and the AA on IP and UP reaches 99.44%, which is 5.90% and 0.54% higher than w/o WSD. In the KSC dataset, AA also increases by 0.21%. Compared with w/o T, the OA of WSD increases by 0.33% and 0.20%, and the AA increase by 0.17% and 0.28%, respectively, on the IP and UP datasets, respectively. Experiments demonstrate that self-distillation can further exploit the latent information of the data and improve classification performance. Compared with w/o Sw, the AA improvement of WSD is more prominent, especially for the IP dataset, which improves by 5.11%. Experiments show that the introduction of weighting based on self-distillation can effectively alleviate the problem of sample imbalance.

F. Influence of Different Number of Training Samples

The number of training samples significantly affects the classification performance of the model. To verify the robustness of the proposed algorithm, experiments are carried out on IP, UP, KSC, and PC with different proportions of training samples. Among them, the training percentages of the IP dataset is 3%, 5%, 8%, 10%, and 15%; the training rate of UP and PC datasets is 0.5%, 1%, 3%, 5%, and 8%, and the learning rate of KSC datasets is 1%, 3%, 5%, 10%, and 15%. Fig. 10 shows the OA for different datasets at different training ratios.

From Fig. 10, it can be seen that OA shows an increasing trend as the training rate increases. For different datasets, the algorithm in this article can reach more than 90% under different training ratios, which can verify its robustness. The PCA-WSD algorithm can reach optimal performance when the amount of data is small. The optimal accuracy can be achieved at 3% for IP and KSC datasets and 1% for UP and PC, alleviating the problem of small samples to some extent.

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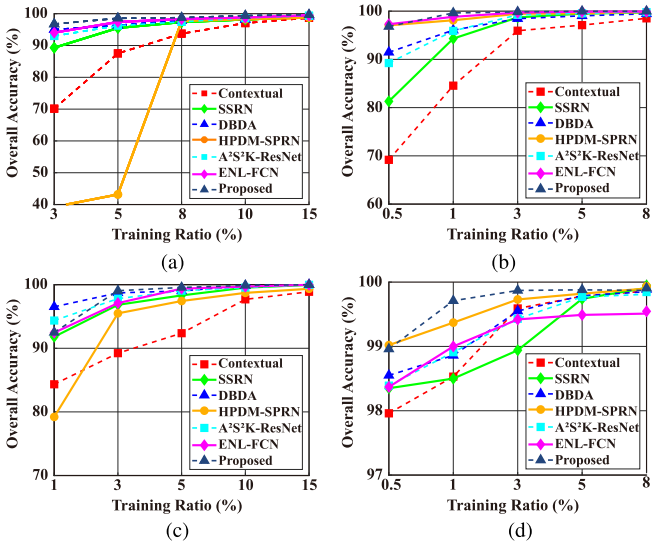


Fig. 10. OA for different datasets at different training ratios. (a) IP. (b) UP. (c) KSC. (d) PC.

TABLE XII
TRAINING TIME AND TESTING TIME OF DIFFERENT ALGORITHMS

Algorithms	Time	IP	UP	KSC
Contextual	Training Time(s)	480.69	195.14	80.34
	Testing Time(s)	46.32	299.61	25.11
SSRN	Training Time(s)	1983.13	620.35	585.96
	Testing Time(s)	22.47	332.29	22.71
DBDA	Training Time(s)	5696.54	742.73	735.81
	Testing Time(s)	145.17	391.17	41.77
HPDN-SPRN	Training Time(s)	442.86	205.99	178.65
	Testing Time(s)	**3.61**	25.25	68.33
A^2S^2K-ResNet	Training Time(s)	4150.58	1190.15	721.47
	Testing Time(s)	91.77	384.17	**18.96**
PCA-WSD	Training Time(s)	166.56	708.78	997.33
	Testing Time(s)	4.63	**24.38**	44.93

G. Time Complexity

To analyze the time complexity, six algorithms are tested on three datasets. The training time and test time of different algorithms are shown in Table XII.

As seen in Table XII, the PCA-WSD spends the shortest testing time on the UP dataset. The time spent on the IP dataset is second only to the HPDM-SPRN algorithm. A^2S^2K-ResNet is the best in the KSC dataset. It can be seen that the testing time of the PCA-WSD algorithm is relatively short, which is related to its input on the whole image. There is a large overlap between adjacent patches, which inevitably increases the computational complexity. PCA-WSD feeds the whole image directly into the network, avoiding repeated calculation.

H. Visualize

To further demonstrate the separability of the features extracted by the PCA-WSD, the t-distributed Stochastic Neighbor Embedding (t-SNE) algorithm [46] is used to visualize the features before and after model's processing. The feature visualization diagram is shown in Fig. 11.

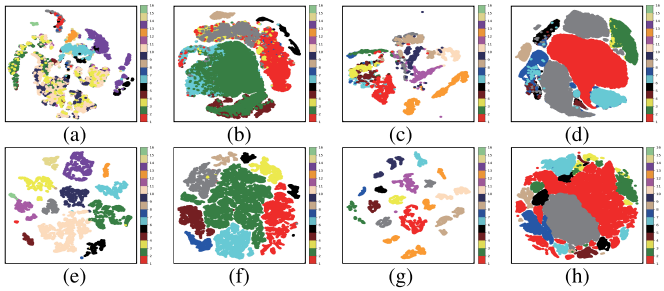


Fig. 11. Feature visualization via t-SNE on four datasets. (a) IP. (b) UP. (c) KSC. (d) PC. (e) IP. (f) UP. (g) KSC. (h) PC.

Fig. 11(a)–(d) shows the original feature distributions of the IP, UP, KSC, and PC datasets, and Fig. 11(e)–(h) shows the feature distributions after the PCA-WSD model, where different colors represent different classes. When the points of the same color are closer and different colors are farther, the extracted features are more discriminative. By comparing the top and bottom rows of Fig. 11, it can be found that the features processed by the PCA-WSD model can make the same class closer to each other compared with the original data features. That means the PCA-WSD model can make the extracted features more discriminative by reducing the intraclass distance and increasing the interclass distance.

IV. CONCLUSION

This article proposes PCA-WSD to address insufficient feature extraction caused by lacking joint attention to spectral–spatial features and ignoring imbalanced samples. PCA-WSD mainly contains SPCS, SPCE, and WSD. SPCS fully uses the advantages of spatial pyramid pooling and coordinated attention to aggregate both local and global features, which significantly enriches the spatial features. Then, diverse aggregated spatial features are embedded into SPCE to realize the joint attention of spectral–spatial features. Finally, WSD is used to capture further the latent information in HSI, which also effectively alleviates the problem of sample imbalance. The algorithm is compared with the seven most advanced algorithms on four public benchmark datasets. Moreover, experimental results show that the PCA-WSD algorithm outperforms other algorithms in qualitative and quantitative metrics, where OA reaches 99.36% and 99.62% on IP and UP datasets. Furthermore, compared with the suboptimal results of the comparative algorithms, AA improves by 1.8% and 1.63% on IP and UP datasets, respectively. The use of PCAM increases OA and AA from 98.91% and 98.85% to 99.36% and 99.44%, respectively, indicating that PCAM is effective for joint attention to spectral–spatial features. Especially when the data is imbalanced, the PCA-WSD algorithm is more effective. OA improves by 5.90% and 0.5% for the IP and UP datasets over the algorithm without WSD, respectively.

The algorithm still needs to be improved. For example, in self-distillation, we only use the teacher's deepest semantic features to guide the student network. However, shallow features also contain a lot of helpful information for classification.

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After that, we will consider combining different hierarchical features in the model to achieve a more accurate classification.

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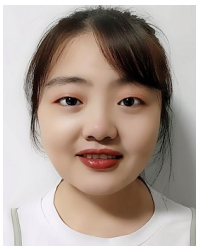
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